**Game Theory in Pharmaceutical Pricing Analysis**

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**1. Static Single-Round Analysis**

**Paper Limitation:** Only modeled one-off bidding interactions

**Enhancement:**

- Added multi-round simulation tracking historical performance
- Implemented dynamic strategy adaptation across rounds

  - Example: `profit_history` and `strategy_history` attributes track evolution

**2. Oversimplified Agent Behavior**

**Paper Limitation**: Companies followed rigid game-theoretic rules without behavioral nuances

**Enhancement:**

- Added risk-aversion parameters (`risk_aversion=0.5`)
- Implemented adaptive follower strategies (margin adjustments based on bid outcomes)
- Leader strategy now considers:

```python
price = price * (1 - 0.1 * self.risk_aversion)  # Risk-adjusted pricing
```

**3. Deterministic Market Modeling**

**Paper Limitation:** Perfectly predictable demand and allocations

**Enhancement:**

- Introduced stochastic demand fluctuations:

```python
base_demand * (1 + (np.random.random() - 0.5) * 0.05)  # ±5% variation
```

- Probabilistic bid acceptance based on risk profiles

**4. Threshold Implementation Gap**

**Paper Limitation**: Proposition 1 thresholds mentioned but not operationalized

**Enhancement:**

- Fully implemented threshold calculations:

```python
threshold1 = (k*(k + 2*self.cost) - A + k*np.sqrt((A - k*self.cost)**2 + F))/(2*k)
threshold2 = (k*(k + 2*self.cost) - A - k*np.sqrt((A - k*self.cost)**2 + F))/(2*k)
```

- Dynamic strategy switching based on real-time thresholds

**5. Lack of Strategic Feedback**

**Paper Limitation:** No learning from past outcomes

**Enhancement:**

- Follower companies adjust margins based on bid history:

```python
if last_profit == 0:  # Lost last bid
   margin = max(0.05, previous_margin - 0.05)
else:  # Won last bid
```

margin = min(0.3, previous_margin + 0.02)

```

**6. Limited Performance Metrics**

**Paper Limitation:** Focused only on immediate profit comparisons

**Enhancement:**

- Added comprehensive metrics:

```python

wins = sum(1 for h in company_history if h['wins_bid'])

avg_profit = np.mean([h['profit'] for h in company_history])

```

- Win rate calculations and market share tracking

**Key Theoretical Alignment:**

The enhanced simulation now properly implements the paper's Proposition 1 thresholds while **adding dynamic elements that maintain fidelity to the original game-theoretic framework.** The risk-aversion extension builds upon the paper's Nash equilibrium concepts by **introducing bounded rationality.**
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